Freeze-Thaw Evaluation of Li-Ion Cells of Varying Form Factor for Application in Mars Landers & Rovers

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Abstract

Extreme temperature fluctuations on the surface of Mars have been shown to accelerate degradation of lithium-ion batteries (LIBs) used for power storage in rovers and landers, motivating improved battery design for extended operational lifetimes. This study examines how the form factor of cylindrical LIBs influences degradation rate, characterized by capacity loss, to inform optimization of battery sizing. Cells of varying form factor were subjected to a thermal cycle replicating the Martian diurnal temperature cycle (DTC), followed by repeated charge-discharge cycling to simulate operational use. Experimental results showed that larger form factor cells suffered more significant degradation due to freeze-thaw exposure, exhibiting a greater divergence in capacity loss relative to control cells. This behavior is attributed to the prominence of thermal gradients and internal heating in larger cells. These findings suggest that minimizing battery size can enhance longevity in environments characterized by extreme thermal fluctuations.

1. Introduction

Since 1997, landers and rovers have been exploring the surface of Mars in search of evidence of life and to better understand the planet's climate and geology. An extreme condition on the surface of Mars that hinders such exploration is the high thermal variability over each diurnal cycle due to a weak atmosphere that is unable to retain heat. The Curiosity rover, located close to the Martian equator and equipped with its Ground Temperature Sensor (GTS) **[1]**, detected an average daily minimum temperature of around -70°C (203 K) and an average daily maximum temperature of around 0°C (273 K) **[2]**. Since each Martian solar day is just over 24 hours long, the temperature at the planet's surface fluctuates between these thermal extremes with high frequency. These extreme temperature swings can have adverse effects on critical components of rovers, most notably their battery systems. Continuous freeze-thaw cycling has been shown to accelerate the degradation of Li cells **[3]** similar to those present on rovers leading to premature failure of the rover. Missions to Mars demand a great deal of time and resources. Therefore, it is vital that these rovers and landers are optimized to operate for as long as possible and carry out their missions successfully. A possible approach to this would be identifying how the accelerated degradation of Li cells during freeze-thaw cycling can be lessened to maximize their lifetime.

Previous research **[3]** has studied the deformation and degradation under repeated freeze-thaw thermal cycling of commercial 18650 Li-ion batteries (LIBs) which are similar to those utilized in spacecraft. This study showed that, when subjected to high charge-rates (e.g. 4C), the effect of freeze-thaw stresses was exacerbated. Synchrotron X-ray computed tomography (XCT) scans of the freeze-thaw cycled cells revealed severe interfacial delamination and increased jelly-roll buckling in comparison to regular cells. These mechanical deformations grew in severity when subjected to repeated charging and discharging. Consequently, the internal resistances of the freeze-thawed cells rose and their charge capacities fell at a greater rate than cells maintained at room temperature (20°C), indicating faster degradation. A key takeaway from this study is that freeze-thaw cycling alone does not produce detectable mechanical damage to the internal structure of these cells. However, thermal cycling in combination with repeated charging and discharging significantly worsened existing damages leading to increased degradation.

In addition to freeze-thaw effects on batteries, previous research has also explored the impacts of form factor on battery degradation. A study by Khange *et al.* **[4]** has shown that the aging behaviour of cylindrical LIBs is influenced by their form factor (physical dimensions) under normal laboratory conditions. Cells of varying form factor were cycled at 25°C with a charge-rate of 1C. Larger cells have poorer surface-to-volume ratios leading to less heat dissipation which in turn accelerates Solid Electrolyte Interface (SEI) formation on the anode. In agreement with this theory, the investigation carried out in **[4]** showed that the thickness of the SEI layer and lithium plating increased at a greater rate in cells of higher form factor. As a result, the relative capacity loss was more significant in these higher form factor cells.

Since freeze-thaw cycling worsens internal damages and LIB form factor corresponds to its degradation rate, it is likely that form factor would influence the accelerated degradation of these cells under freeze-thaw cycling. The Chen *et al.* **[3]** study characterized the accelerated degradation of 18650 cells when subjected to thermal cycling. However, it does not study the accelerated degradation of any other cylindrical cell types to identify whether form factor could influence to what degree thermal cycling contributes to cell degradation. This study aims to experimentally investigate the impact of form factor on the accelerated degradation of LIBs subjected to freeze-thaw cycling utilizing cylindrical Li-ion cells of varying physical dimensions. In both **[3]** and **[4]**, degradation is caused by internal heating and thermal gradients which are more prominent in larger cells. Therefore, it can be hypothesized that larger form factor LIBs are more likely to degrade at a higher rate when subjected to repeated thermal cycling. Knowledge on how the size of the LIBs undergoing repeated freeze-thaw influences their lifespan would be highly applicable to the design of power storage systems for Mars rovers and landers. It would allow for the optimization of battery designs to improve their lifespan in extreme thermal conditions similar to those found on the surface of Mars.

2. Methodology

Seven cell types of varying form factor and initial capacity were used in this investigation and labelled as shown in Table 1. The form factors investigated were 16340 (16 mm diameter, 34 mm height), 18650 (18 mm diameter, 65 mm height), 21700 (21 mm diameter, 70 mm height) and 26650 (26 mm diameter, 65 mm height). It was observed that the manufacturer rated capacities of these cells differed from the measured capacities even at low (0.5C) charging rates, possibly due to variations in manufacturing quality. Although 26650-A and 26650-B cell types have the same form factor and rated capacity, they differ in manufacturer, quality and average measured capacity.

Label	Form Factor	Diameter (mm)	Height (mm)	Rated Capacity (mAh)	Average Measured Capacity (mAh)
16340	16340	16	34	2800	660
18650-A	18650	18	65	2500	2440
18650-B	18650	18	65	3000	3020
18650-C	18650	18	65	3300	1960
21700	21700	21	70	5000	4960
26650-A	26650	26	65	5000	1890
26650-B	26650	26	65	5000	5030

Table 1: Data for different cell types used in investigation

2.1. Charge-discharge cycling

In Chen *et al.* **[3]**, the LIBs were cycled at a constant C-rate using a constant-current, constant-voltage (CC-CV) charging protocol. The cells were charged at a constant current until a voltage of 4.2 V was reached, after which this constant voltage level was maintained until the current decreased below a cut-off value, followed by discharge back to 2.5 V. This investigation used an identical approach as shown in Figure 1 with a current cut-off value of 0.1 A. The charging-discharging cycles were performed at room temperature (20°C) using a Landt battery cycler CT3002A. Cell degradation was characterized by the reduction in cell capacity over many charge-discharge cycles. Control cells of each type maintained at room temperature (20°C) were repeatedly charged and discharged at a constant C-rate of either 0.5C or 2C to account for degradation during repeated charging cycles. The cells exposed to freeze-thaw conditions were similarly subjected to repeated charging cycles to compare against the control cells and identify signs of increased degradation.

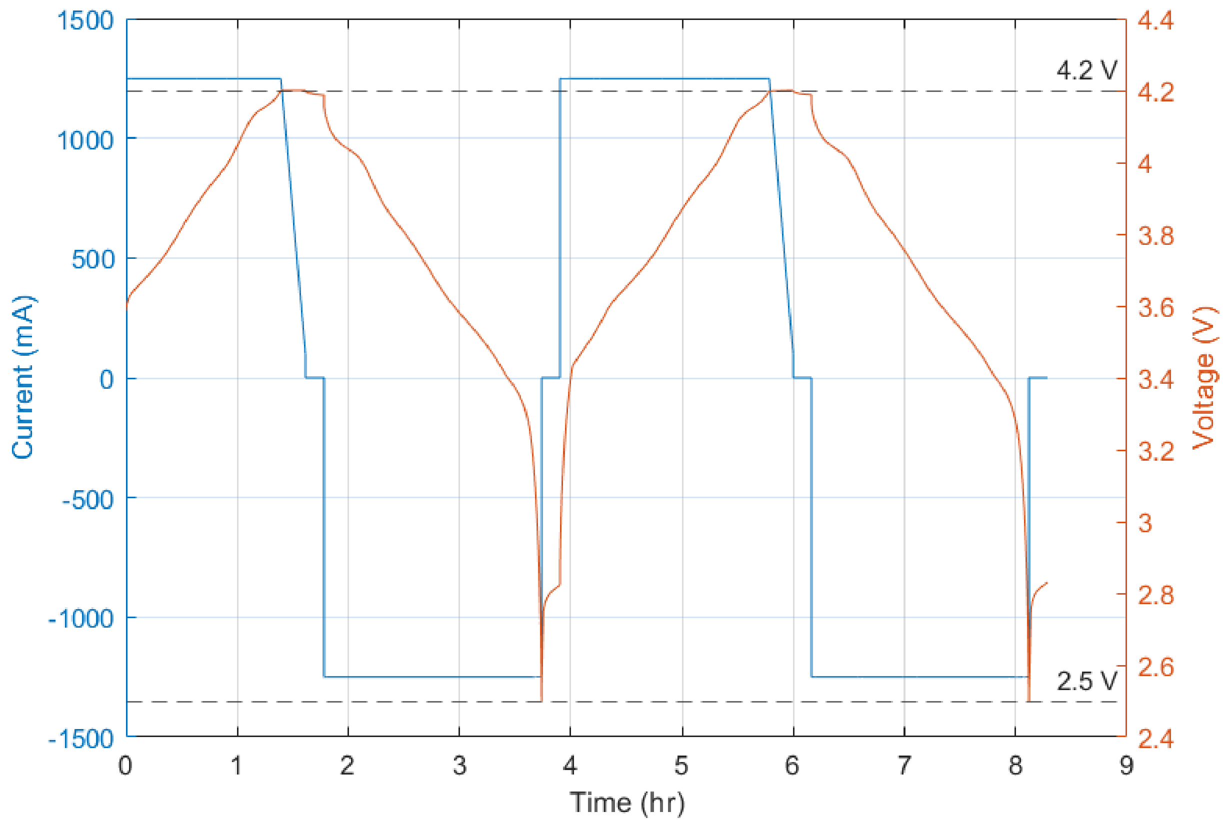

Figure 1: Current and voltage readings during 0.5C charging and discharging of 18650-A cell

2.2. LIB behaviour at low temperatures

Li-ion cells of each cell type were exposed to subzero temperatures representative of daily surface conditions on Mars. This allowed for the observation of LIB behaviour within the temperature range being investigated, specifically how the electrical properties of the cells such as capacity and internal resistance varied with temperature.

Capacity was determined at each temperature by cooling each cell and running charge-discharge cycles at a 0.5C charging rate to calculate the maximum charge that could be supplied to it and the maximum charge that could be extracted from it (within a fixed voltage window) while at that temperature. In order to charge and discharge the batteries while maintaining a constant subzero temperature, cables were set up within the cryogenic cooler that connected the batteries to the Landt battery cycler. It is important to note that these two instruments were physically spaced apart in the laboratory space, so the cables connecting them were required to be roughly 3 meters in length. Therefore, it is likely that the resistance within these cables may have increased the voltage drop at the voltage sensor, leading the battery cycler to sense a greater voltage than is actually present across the battery, impacting the capacity values recorded. However, since the same setup was used at each temperature, the impact on the magnitude of the capacity should be similar, meaning any trends and patterns observed should be valid.

The direct current internal resistance (DCIR) was determined using an experimental method similar to that outlined in **[3]**. The state of charge (SoC) of LIBs of each type was set to 50% at room temperature by charging the cells using CC-CV at 0.5C, allowing them to rest, and then discharging at 0.5C for 1 hour so that approximately half of their maximum charge remained. Afterwards, these batteries underwent a pulse power test in which a single discharge pulse was applied at 1C for 10 seconds. The instantaneous voltage drop recorded (and shown in Figure 2) is equal to the pure Ohmic resistance of the cell, R_0, which is the bulk electrolyte ionic resistance of the cell **[5]**. The pulse power test was repeated at each temperature to compute an average value for the internal resistance.

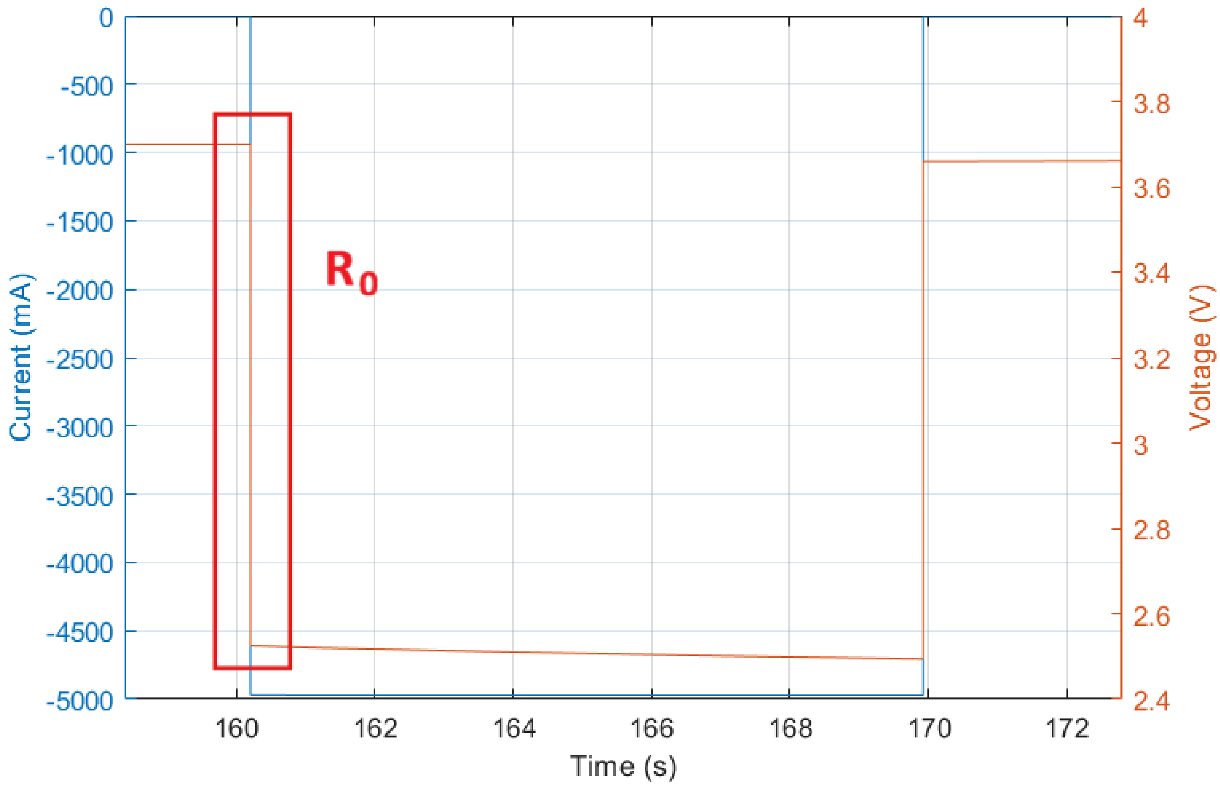

Figure 2: Current and voltage readings during a power pulse test on a 21700 cell (current and voltage values used to determine pure ohmic resistance R_0 are outlined in red)

2.3. Temperature curve for freeze-thaw

The freeze-thaw procedure used in this investigation differs considerably from that used by **[3]**. In their experiment, Chen *et al.* **[3]** froze the LIBs at -150°C for 24 hours and then thawed these cells at room temperature (20°C) for another 24 hours. In contrast, this investigation utilizes a cooling curve that more closely resembles the temperature changes experienced on the surface of Mars.

The Ground Temperature Sensor (GTS) of the Curiosity rover recorded the variation of temperature near the Equator during each diurnal cycle over several hundred sols. Figure 3 shows the ambient air temperature recorded by the GTS during a 13-hour period of the rover's 224th solar day on Mars. Temperature data such as that shown in Figure 3 was obtained through the Planetary Data System (PDS) archive of the National Aeronautics and Space Administration (NASA) **[2]** for multiple solar days, and used to construct a simple temperature curve that approximately resembled the diurnal temperature cycle of Mars (Figure 4 and Table 2). Due to thermal variation across different sols and at various locations near the equator, a high degree of accuracy was unnecessary for constructing the temperature curve using empirical data and it was considered acceptable to approximate the curve.

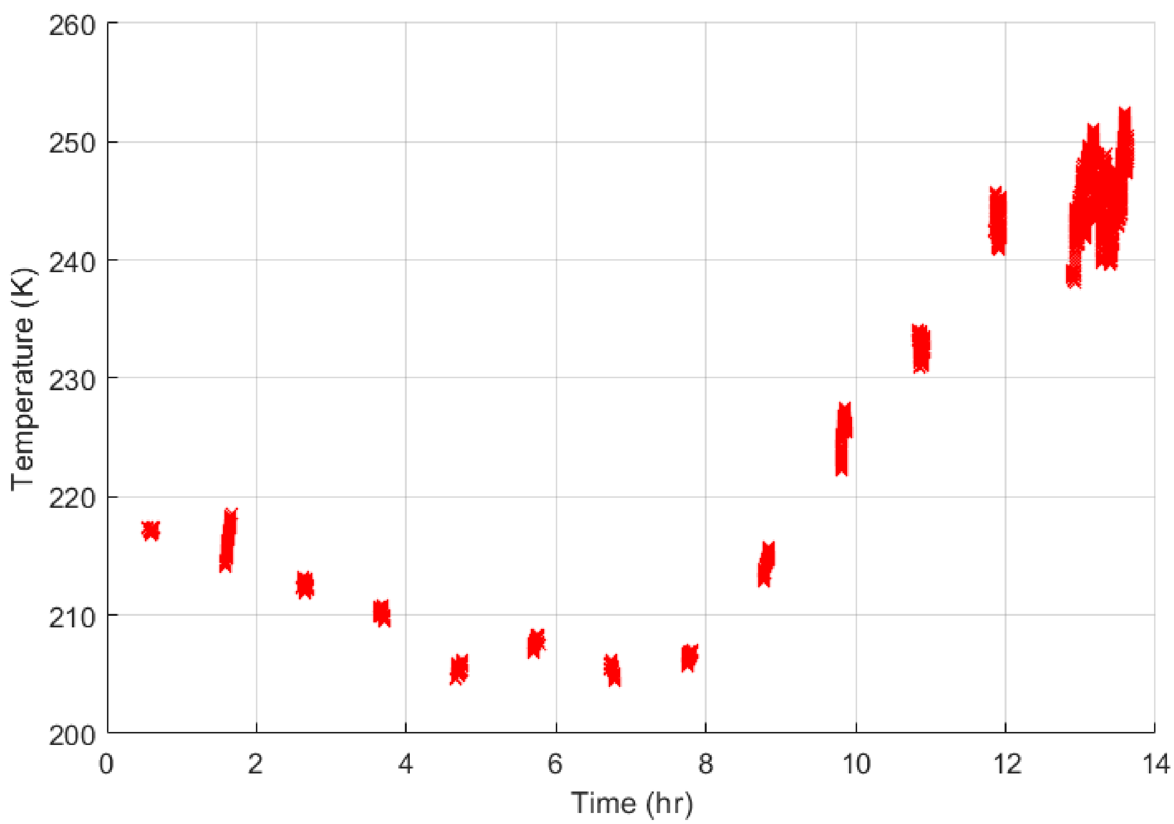

Figure 3: Ambient air temperature during 13-hour period of Curiosity rover's 224th solar day on Mars as recorded by the Ground Temperature Sensor

A cryogenic chamber (MDF-150C118 Qingdao Antech Scientific Co., Ltd.) was used to cool the LIBs to the required temperatures. The State of Charge (SoC) impacts the freezing temperature of the electrolyte so all cells were fully discharged before being placed in the cryogenic chamber. Figure 4 displays the temperature curve over two solar days. It can be observed that the maximum temperature peaks are shorter in duration compared to the minimum temperature peaks. The electrolyte within the LIBs has a freezing temperature of around -25°C, meaning that, during each cycle, the Li-ion cells spend 18 hours in a frozen state and 6 hours in a thawing state. This is in sharp contrast to the 24 hours of freezing followed by 24 hours of thawing utilized in **[3]**. Our study focuses on the accelerated degradation of LIBs when specifically subjected to Mar's extreme temperatures. Therefore, moving forward, this investigation employs the freeze-thaw cycle that follows the temperature curve shown in Figure 4 resembling the Martian diurnal temperature cycle.

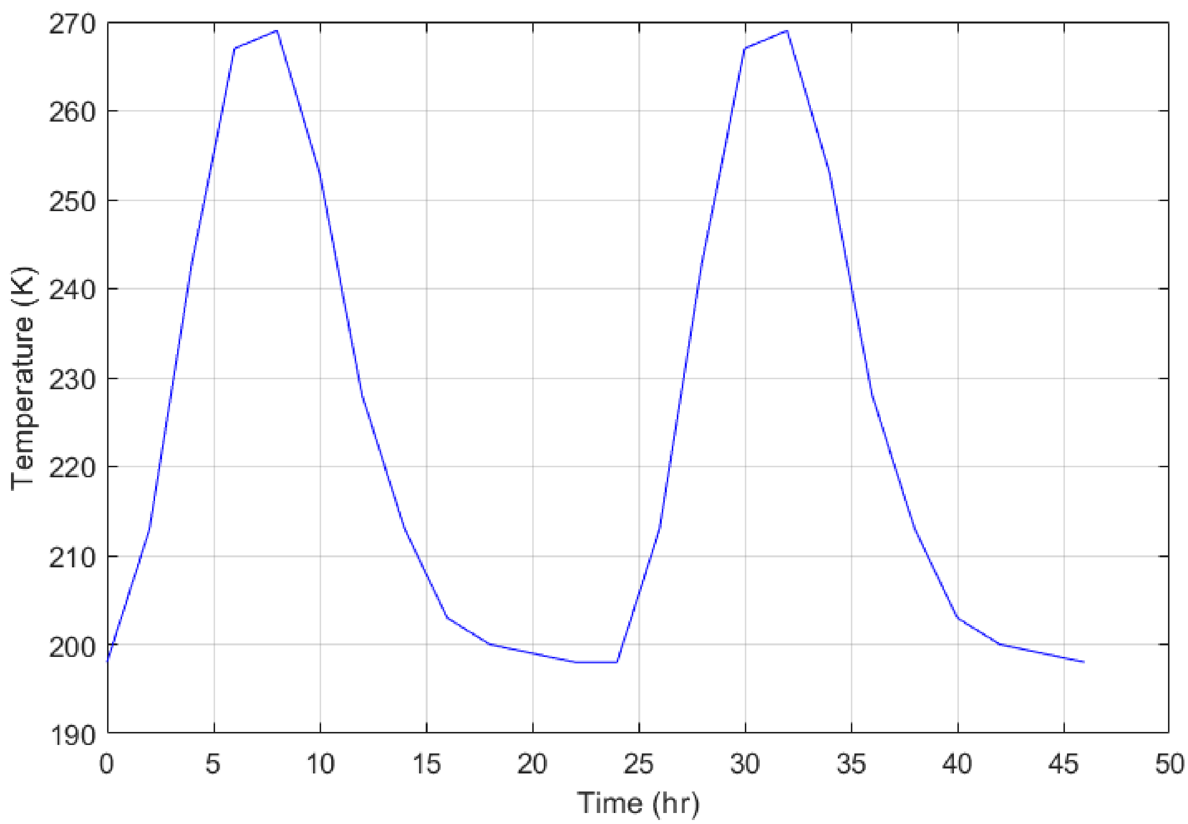

Figure 4: Constructed temperature cycle (shown for two solar days) approximating the diurnal temperature cycle of Mars

Time (hr)	0	2	4	6	8	10	12	14	16	18	20	22
Temp (°C)	-75	-60	-30	-6	-4	-20	-45	-60	-70	-73	-74	-75

Table 2: Temperature values used to approximately replicate the diurnal temperature cycle of Mars

3. Results & Discussion

3.1. Capacity variation with temperature

Figure 5 shows the reduction in cell capacity in mAh for cells of type 18650-A, 18650-C, 21700 and 26650-A. As noted earlier, the high resistance in the cables connecting the batteries in the cryogenic cooler to the Landt battery cycler would have increased the voltage drop across the sensors on the Landt battery cycler, causing it to sense the CC-CV constant voltage of 4.2 V much earlier while the voltage across the actual battery was much lower. Consequently, the batteries were not charged to their full capacity, but since an identical setup was used at all temperatures for all four cells, the impact on the capacity values should be identical, so valid trends can still be observed.

For all four LIBs, the capacity measured at 20°C (equal to room temperature) within the cryogenic cooler was roughly half the expected capacity measured at room temperature outside the cryogenic cooler (without the increased resistance of the connecting cables). Capacity measured at 5°C was lower than at room temperature for all batteries. The degree to which this reduction occurred varied across the four cell types from a 19% reduction in the 18650-A cell and an 80% reduction in the 21700 cell. No concrete correlation was observed between the degree of capacity reduction and other battery characteristics. Similarly, between 5°C and -10°C, the capacity decreased once more for all cells to varying degrees. This is expected at lower temperatures since the viscosity of the electrolyte rises as it prepares to solidify, inhibiting movement of ions and transfer of charge within the cell and limiting power output. At -25°C, the capacity of all the LIBs dropped to approximately 0 mAh and remained at this level for subsequent lower temperatures, indicating that the electrolyte had solidified to the extent that no net charge transfer was possible. This is also consistent with our earlier approximation of -25°C as the freezing temperature of the electrolyte. Therefore, the battery can be identified as being in the “frozen state” of the freeze-thaw cycle around -25°C.

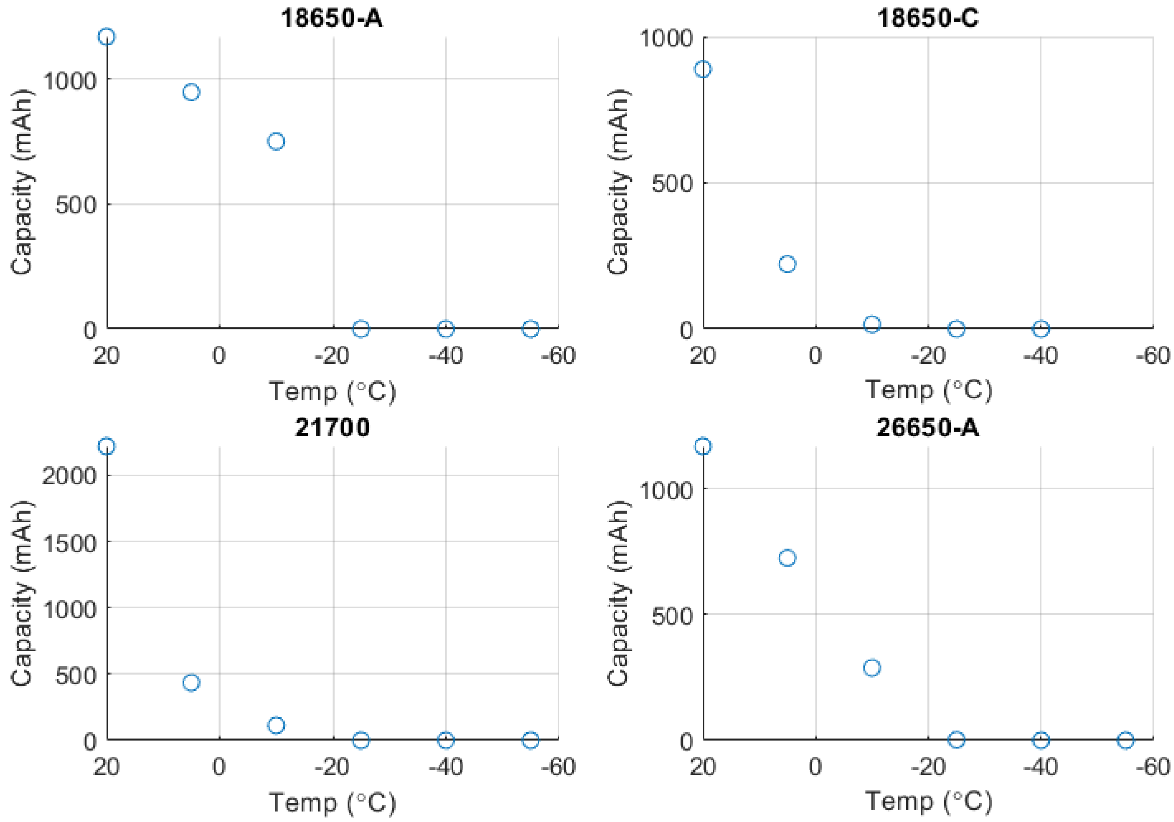

Figure 5: Cell capacity measured at various temperatures

3.2. Internal Resistance variation with temperature

The growth in internal resistance with decreasing temperature was analyzed using the experimental results displayed in Figure 6. Similar to the above experiment investigating capacity-temperature variation, this experiment also faced the challenge of collecting measurements through the Landt cyler while the cells were at a fixed temperature within the cryogenic chamber. As a result, the resistance within the cables connecting the cycler to the cooler would have contributed to the resistance detected during the pulse power test. However, because the wires and battery were connected in series with each other, the cable resistance would have simply added to the internal resistance of the battery when measured. Since the same setup was used at all temperatures, the same cable resistance would have been added to all battery internal resistance values, so the relative change in internal resistance between temperatures remains accurate.

The internal resistance values of all the cells were in the range of 20 to 70 mΩ at room temperature (20°C). This resistance increased when the temperature was decreased to 5°C, and continued to increase at subsequent temperatures -10°C and then -25°C. As seen in Figure 6, the growth in resistance between 20°C and -25°C was roughly linear. As temperature decreases, the ions within the electrolyte lose energy which impedes their movement and the transfer of charge within the cell. As the temperature decreases further, ion movement is further inhibited, contributing to the growth in cell resistance. Beyond -25°C, however, the internal resistance increases at a much steeper rate reaching several thousand mΩ by -40°C. The Landt battery cycler was unable to measure resistance values at temperatures beyond -40°C for any of the cells as the resistance was too high and the maximum voltage threshold for safe operation was met. This is likely because the freezing point of the electrolyte had been reached, at which point the electrolyte had solidified, preventing any further motion of ions and, by extension, the flow of current.

A weak correlation was observed in the data where cells with larger form factors exhibited greater internal resistance at low temperatures in comparison to some of the smaller cells as visible on Figure 6. This observation was unexpected as larger batteries have greater cross-sectional area, facilitating better flow of ions within the cell, thereby reducing its internal resistance. An exception to the observed trend was 18650-C which had a smaller form factor but still exhibited high internal resistance.

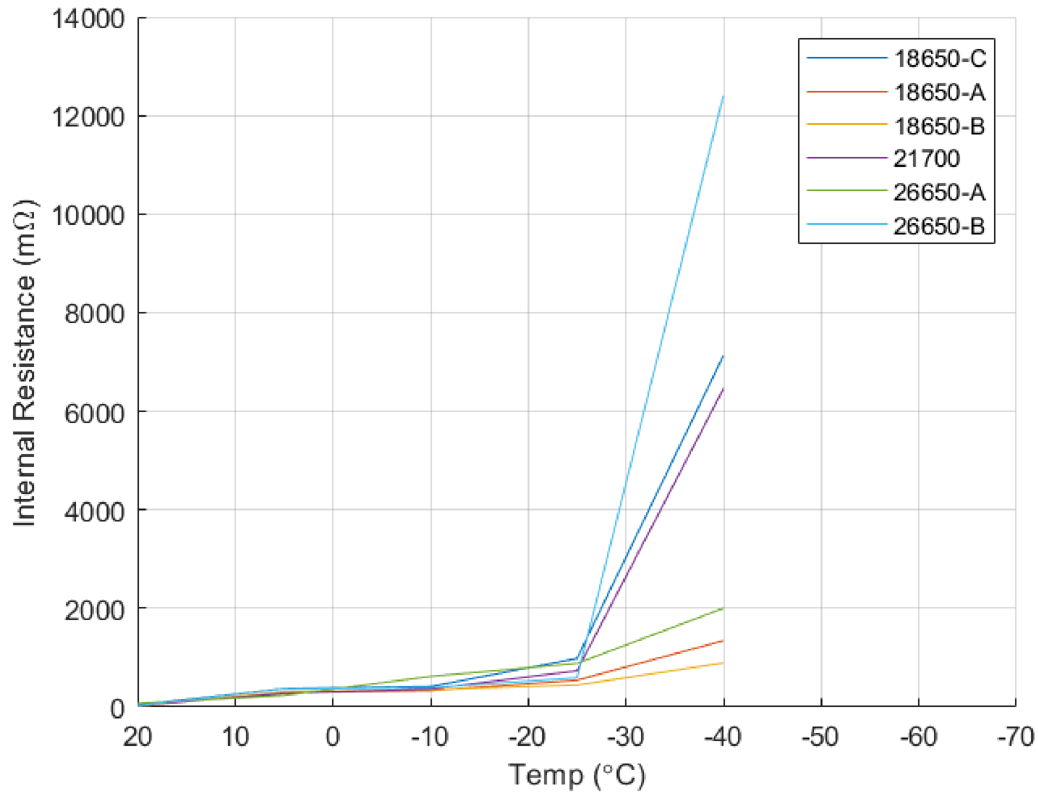

Figure 6: Internal resistance measured at various temperatures

The data from the 16340 cells were omitted from Figure 6 due to a procedural error which impacted the measured values. Cells of this type were much too small for the battery cases available in the laboratory so a small iron conductor was placed in series with the battery to allow it to fit into the battery cases. This would have had a negligible impact on the capacity measurements since no charge could be stored within the conductor. However, the resistance within the iron conductor likely added to the internal resistance value measured by the battery cycler. It is worth noting here that the entire battery case was cooled inside the cryogenic chamber, so the resistance of the conductor may have also increased at lower temperatures, disrupting the overall internal resistance measurements and making the 16340 dataset unreliable.

3.3. Capacity variation over charging cycles

The investigation aims to probe the impact on cell degradation caused by repeated freeze-thaw exposure. The degradation of Li-ion cells is characterized by the reduction in its capacity due to use which is replicated in this experiment through repeated charging and discharging of the cells. To understand the impact of repeated freeze-thaw exposure on battery degradation, it is necessary to first examine the degradation of these cells when maintained at room temperature to serve as a basis for comparison. Accordingly, several control cells were subjected to 50 charge-discharge cycles and the results of this experimentation are presented in Figure 7a & 7b.

Cells degrade with use due to cumulative irreversible chemical and physical changes within the cell that cause permanent damage. Growth of SEI layers, increased lithium plating and decomposition of the electrolyte are three such changes that contribute significantly to capacity loss. This can be seen in Figure 7 where the discharge capacity (the energy in mAh that could be extracted from the battery) reduces over the 50 charge-discharge cycles for all tested cells as internal damages accumulate. Charging and discharging at higher C-rates can exacerbate these damages. Higher current flow within a cell can lead to heat generation through internal friction which can further damage internal components of the cell. Similarly, lithium plating on the anode is increased at higher currents as ions that are unable to enter the anode fast enough deposit as metallic lithium on its surface contributing to capacity loss. Table 3 compares the percentage capacity of each battery after 50 charge-discharge cycles when cycled at a low C-rate (0.5C) and at a high C-rate (2C). As expected, a greater loss in cell capacity can be observed when cycled at 2C compared to 0.5C for all batteries tested.

Surprisingly, this increase in capacity loss with charging rate was most significant for the 16340 and 18650-C cells. In the study conducted by Khange *et al.* **[4]** outlined earlier, it was found that cells of higher form factor experienced greater SEI formation and lithium plating which resulted in greater degradation and capacity loss. This pattern is not clearly discernable from the data presented in Figure 7 as there does not appear to be a correlation between form factor and capacity reduction rate. This discrepancy is likely due to the difference in number of charge-discharge cycles completed in this investigation (50 cycles) relative to that completed in **[4]** (600 cycles). In their study, the variation in capacity loss only becomes discernible beyond 200 cycles. Therefore, it is possible that the influence of form factor on degradation rate at room temperature only becomes substantial after a large number of charge-discharge cycles, justifying why it may not have had a discernible effect after only 50 cycles. It should also be noted that, even after 600 cycles, the difference between the relative capacity of 18650 cells and 26650 cells was only 1-2%. This could indicate that the effect of form factor on degradation rate may be trumped by other factors in this case such as manufacturing quality.

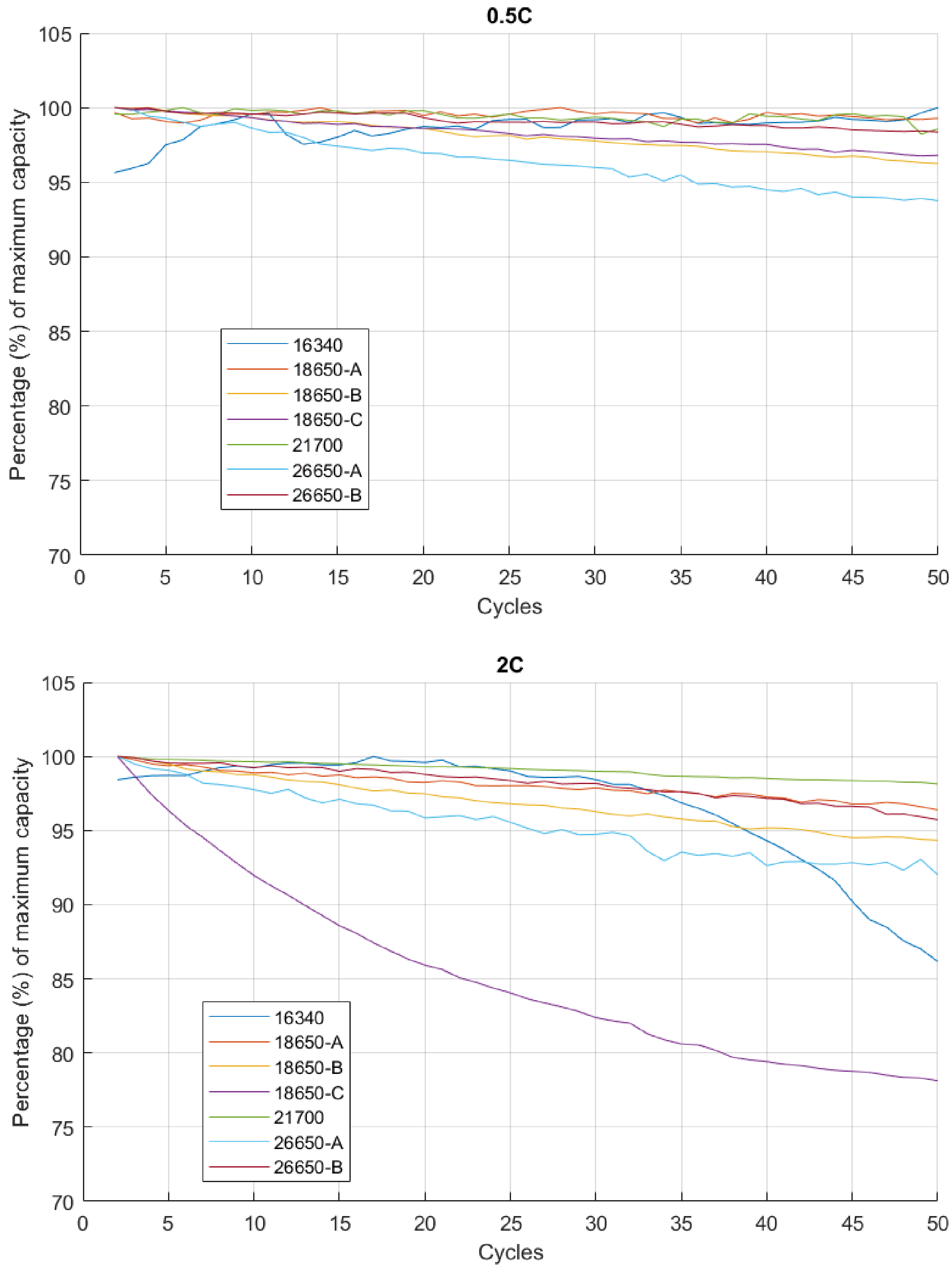

Figure 7: [top] a) Change in capacity over 50 charging cycles at 0.5C; [bottom] b) Change in capacity over 50 charging cycles at 2C

Battery Type	16340	18650-A	18650-B	18650-C	21700	26650-A	26650-B
50 cycles at 0.5C	100%	99.3%	96.3%	96.8%	98.6%	93.8%	98.4%
50 cycles at 2C	86.2%	96.4%	94.3%	78.1%	98.1%	92.0%	95.7%

Table 3: Remaining capacity after 50 cycles at 0.5C and 2C

3.4. Impact of repeated freeze-thaw cycling

As discussed earlier, to replicate the thermal conditions on the surface of Mars, a temperature curve was constructed using empirical data to match the Martian diurnal temperature cycle (DTC). The batteries were subjected to 14 consecutive DTCs, left to rest for 15 hours and then cycled at 2C for 80 charge-discharge cycles. Figure 8 shows the percentage capacity loss over 80 cycles for cells subjected to the freeze-thaw conditions (blue) and control cells maintained at room temperature (orange) for each battery type. Table 4 compares the remaining percentage capacity of each cell after 80 charging cycles.

For 18650-A and 18650-C batteries, there is no observable difference between the capacity lost by cells subjected to the thermal cycles and the cells maintained at room temperature. This is within reason because in **[3]**, 18650 cells cycled at 4C for 1000 charge-dicharge cycles exhibited a 13% difference in the capacity loss by the freeze-thawed cells and control cells. When cycled at 2C for only 80 cycles, it is plausible that the damage from thermal-cycling had less effect. For the 16340 and 18650-B cells, the cells subjected to the thermal cycles appeared to have lost slightly less (2.7% and 1.9% respectively) capacity than the control cells. It is possible that this could be individual cell variation as freeze-thaw exposure is unlikely to improve cell capacity. For the larger form factor cells, the thermal-cycled cells experienced greater capacity loss than their controlled counterparts, with the 26650 cells showing a greater divergence than the 21700 cells.

Cells of larger form factor appear to have suffered a greater reduction in capacity due to thermal-cycling relative to cells of smaller form factor. Since reduction in capacity is characteristic of cell degradation, these results indicate that increased form factor correlates to increased cell degradation when subjected to freeze-thaw conditions. This is consistent with the hypothesis put forward earlier that larger form factor cells will experience greater degradation since it is strongly influenced by internal heating and thermal gradients that are more prominent in larger cells. Therefore, it is reasonable to conclude that as identified in **[3]**, LIBs do experience greater degradation when subjected to repeated freeze-thaw conditions, but, as shown through this investigation, larger cell form factor can worsen this accelerated degradation.

Future research could investigate how this relationship between form factor and degradation rate evolves if the cells are subjected to over 1000 charging cycles. Similarly, further exposure to freeze-thaw conditions beyond 14 DTC can also be studied to investigate the effects of prolonged exposure to these conditions. Both paths for potential future research would contribute to our understanding of how thermal-cycling induced damage influences the long-term health of batteries used during Mars surface missions.

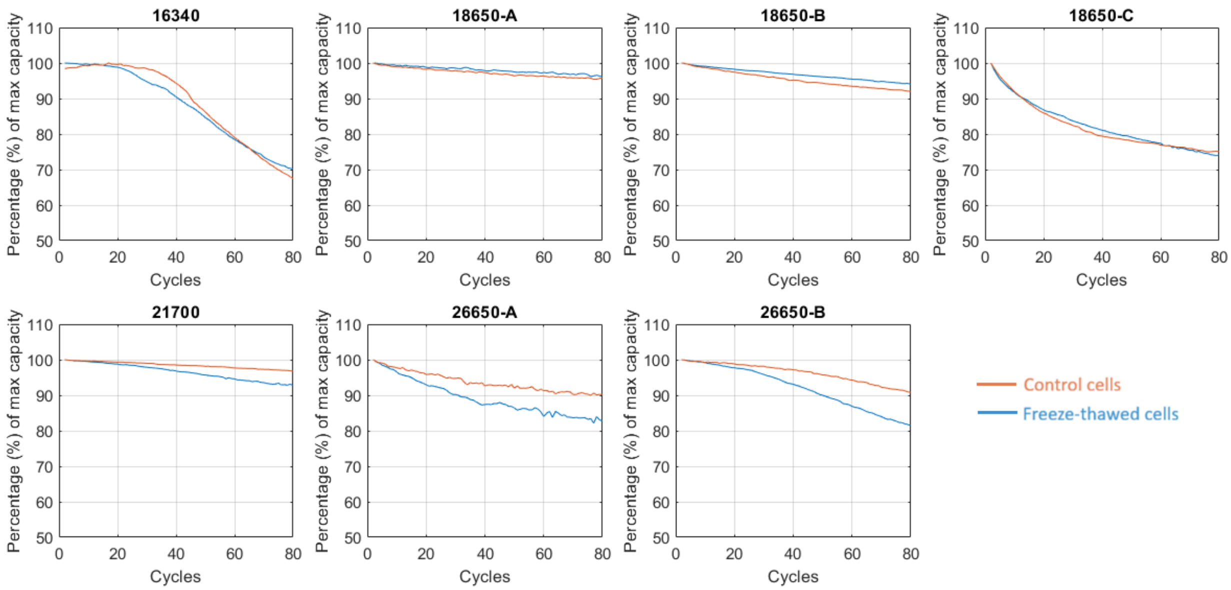

Figure 8: Comparison of capacity of freeze-thawed and control cells after 80 cycles at 2C

Battery Type	16340	18650-A	18650-B	18650-C	21700	26650-A	26650-B
Control (20°C)	67.5%	96.2%	92.1%	75.1%	96.8%	90.2%	90.8%
Freeze-thawed	70.2%	95.7%	94.0%	74.1%	92.7%	82.8%	81.2%

Table 4: Remaining capacity after 80 cycles at 2C for freeze-thawed and control cells

4. Conclusion

This study evaluates the influence of form factor on the accelerated degradation of cylindrical lithium-ion batteries when exposed to repeated freeze-thaw cycling similar to conditions experienced on the surface of Mars. Experimentally-determined relationships between capacity, internal resistance and temperature helped verify that the cells entered a “frozen state” around -25°C. A temperature curve was constructed using empirical data to replicate the diurnal temperatures cycle and Li-ion cells were exposed to it in a laboratory setting using a cryogenic chamber. Cells exposed to 14 consecutive thermal cycles were subsequently subjected to charge-discharge cycles at 2C. As expected, when compared to control cells maintained at room temperature, these cells exhibited increased degradation in the form of greater relative capacity loss after 80 cycles. While repeated freeze-thaw events have not been observed to cause direct damage to cells, they exacerbate damages caused due to charging cycles, thereby accelerating the degradation of the battery. The experimental results also agreed with the proposed correlation between form factor and degradation rate where larger cells appeared to suffer greater reduction in capacity. 26650 and 21700 cells saw a greater difference between the capacity loss of freeze-thawed cells and control cells in comparison to 18650 and 16340 cells. This is likely due to internal heating and thermal gradients that elevate cell degradation and have a greater effect in larger cells. Therefore, as evidenced by these experimental results, it can be concluded that Li-ion cells of larger form factor are more impacted by freeze-thaw events compared to their smaller counterparts, leading to increased degradation in these larger cells. This understanding could be highly applicable to the design of energy storage systems for Mars rovers and landers. This finding would signify that the lifespan of batteries exposed to such extreme thermal fluctuations could be optimized by regulating and minimizing their physical dimensions. Overall, this study contributes to the growing body of research on battery behaviour in extreme thermal environments that guide the development of more efficient battery designs for space applications.

5. References

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6. Acknowledgements

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